Synaptic Plasticity as Short-Term Memory

A mechanism-level briefing on outer-product Hebbian writes, fixed recurrent state, and the interference boundary exposed by the Synaptic Memory Lab.

PROBLEM AND DESIGN PRESSURE

Autoregressive Transformers preserve exact access to prior context by retaining key/value pairs for previously generated tokens. For a head width d and sequence length T, the decoder-side cache grows as $O(Td)$ per layer and head family. This creates a memory wall for long-horizon generation because each additional token adds persistent state that must be addressed, moved, and eventually evicted. The design pressure is to retain useful associations without retaining every historical record.

MECHANISM

Linear attention exposes one route by regrouping query–key–value computation into a recurrent fast-weight state. Given a key k_t, target/value v_t, decay λ, and plasticity η, the state update is:

$$M_t = \lambda M_{t-1} + \eta(k_t \otimes v_t) \qquad y_t = M_t q_t$$

The outer product writes a rank-one association into a matrix-shaped state; the query retrieves a superposition of written values. With fixed width, M occupies $O(d^2)$ state regardless of the number of presented items. This is the same algebraic shape as a Hebbian rule, $\Delta W_{ij} = \eta x_i y_j$: co-active pre- and post-synaptic patterns strengthen their connection. The fixed-state guarantee changes the failure mode. If $q_i^T k_j$ is large for the wrong pair, the read contains a cross-term proportional to $q_i^T k_j v_j$. Increasing load or key collinearity therefore produces associative drift rather than a clean miss.

ROLE OF DRAGON HATCHLING

The Dragon Hatchling (BDH) proposal places memory and reasoning in a biologically inspired network of locally interacting neuron particles. Its GPU-friendly formulation connects sparse non-negative activity, low-rank transformations, and linear-attention-compatible recurrence. The relevant conceptual point is that synapses can be treated as an active working-memory substrate instead of a passive parameter store. A sparse firing rule $\varphi(x)=\max(0,x-\theta)$ can reduce weak, sign-canceling activity before a write or read. At roughly five percent active units, sparse firing may improve signal-to-noise in a local write, but it does not remove finite-capacity interference. BDH-GPU should not be collapsed into a Mamba-style selective state-space model: both support recurrent inference, but their state semantics and architectural assumptions differ.

ARCHITECTURAL COMPARISON

Architecture	State footprint	Per-token compute	Update	Capacity limit	GPU note
Transformer / softmax KV	O(Td)	O(Td)	Append K/V	Context + hardware budget	Optimized kernels; traffic grows with T
Mamba / SSM	O(d) to O(dN)	Linear in state width	Input-dependent transition / scan	Compression + state dimension	Strong scan kernels; implementation-dependent
Linear attention / fast weights	$O(d^2)$	$O(d^2)$	Decay + outer product	Rank + key overlap	Parallelizable; matrix bandwidth matters
BDH / BDH-CQ	Architecture-dependent sparse state	Sparse graph + low-rank path	Firing with synaptic updates	Finite synapses + interference	GPU-friendly formulation reported by developers

EVIDENCE DISCIPLINE

The BDH preprint reports developer-run scaling and task experiments; those remain developer-reported unless independently reproduced. A reproduction matches the protocol and metrics in a separate implementation. A production deployment establishes operational use, not benchmark validity. This project claims none of those outcomes: its cosine similarity and MSE come from a deterministic browser toy engine so the interference mechanism can be inspected directly.

LIMITATIONS AND NEXT STEPS

The fixed matrix is a finite-rank information bottleneck. Orthogonal patterns coexist only within the effective dimension and noise margin; non-orthogonal keys create cross-talk earlier. Thresholding may suppress weak noise while deleting low-amplitude information, and decay trades persistence for freshness. The abstraction omits learned projections, normalization, optimization dynamics, hardware tiling, and the full BDH graph. A next experiment should sweep key coherence and load independently over randomized seeds, compare dense and thresholded reads, and study how fast synaptic state could be consolidated into slow weights without freezing errors.

Primary literature: Kosowski et al., arXiv:2509.26507 (2025); Sun et al., arXiv:2307.08621 (2023); Yang et al., arXiv:2312.06635 (2023/2024); Tyulmankov et al., *Neuron* (2022). Miconi, arXiv:2107.01729, is an implementation bridge for Hebbian learning with gradients.